R-107, First Floor, Metro Power, Vijay Nagar, Indore (M.P.) 4520101

Certificate Course in Generative AI, LLM Applications & AI Agents

A practical curriculum that takes learners from using GenAI effectively to building document assistants, AI workflows, and tool-using agents.

Course structure

Component	Proposed structure
Level	Foundation to intermediate application development
Duration	**90 instructor-led hours + 30 hours of independent practice and project work**
Schedule	15 weeks × 6 hours per week
Teaching approach	Approximately 30% concepts and 70% demonstrations, labs, and projects
Suitable for	Students, analysts, developers, trainers, and working professionals
Prerequisites	Basic Python: variables, collections, conditions, loops, and functions
Optional bridge	12 additional hours of Python for learners without programming experience

Learning outcomes

By the end of the course, learners should be able to:

- Explain how LLMs generate responses and where they can fail.
- Design prompts and evaluate their effectiveness.
- Develop Python applications using LLM APIs.
- Build retrieval-augmented generation (RAG) applications using private documents.
- Create AI agents with validated tools and controlled actions.
- Compare prompting, RAG, and fine-tuning for a business requirement.
- Test and deploy a GenAI application with appropriate privacy, reliability, and cost controls.

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Module-wise curriculum

Module	Hours	Topics	Practical outcome
1. Foundations of Generative AI	4	AI, ML, deep learning, and GenAI; generative versus predictive models; text, image, audio, and video generation; foundation models; use cases and limitations; hallucinations	Map business problems to suitable GenAI approaches
2. Prompt Engineering and Context Design	6	Instructions, context, examples, constraints, and output formats; zero-shot and few-shot prompting; prompt templates; task decomposition; structured outputs; context selection; verification	Build and test a reusable prompt collection
3. Python and API Development Essentials	8	Python refresher; environments and packages; JSON; HTTP requests; API authentication; environment variables; exception handling; Git basics; reading text and CSV files	Build a Python application that calls an API and handles errors
4. NLP, Transformers, and LLM Fundamentals	8	Tokenization; embeddings; semantic similarity; attention and Q/K/V intuition; positional information; encoder, decoder, and encoder–decoder architectures; next-token prediction; pretraining and instruction tuning; context windows	Explore tokenization, embedding similarity, and a small attention example
5. Building LLM Applications	6	Model selection; messages and conversation history; generation settings where supported; streaming; structured response validation; retries; rate limits; token usage and cost tracking	Create a conversational assistant with validated outputs

Module	Hours	Topics	Practical outcome
6. Embeddings and Vector Search	6	Document loading; text cleaning; chunking; metadata; embedding models; similarity metrics; vector indexes; metadata filtering; semantic versus keyword search	Build a semantic search application
7. Retrieval-Augmented Generation	10	Ingestion and retrieval pipelines; prompt construction from retrieved context; citations; hybrid search; reranking; conversational retrieval; document updates; handling missing evidence	Build a PDF/document assistant that cites its sources
8. Multimodal Generative AI	6	Image understanding and generation; document extraction; OCR versus vision approaches; speech transcription and synthesis; video summarization workflows; consent and content rights	Create a document-to-summary or audio-to-action-items workflow
9. AI Agents and Tool Calling	10	Chatbots, workflows, and agents; tool definitions and schemas; tool selection; argument validation; execution loops; state and memory; stopping conditions; human approval; timeouts and failure recovery	Build a course enquiry agent with controlled tool access
10. Model Context Protocol — MCP	4	Host, client, and server roles; tools, resources, and prompts; MCP versus direct API integration; permissions and trust boundaries; connecting an existing server; minimal server demonstration	Connect an assistant to a limited, read-only information tool
11. Model Adaptation and Fine-Tuning	6	Prompting versus RAG versus fine-tuning; training examples; dataset quality; train/validation/test separation; supervised fine-tuning; LoRA and QLoRA concepts; overfitting; evaluation; compute requirements	Prepare a small dataset and complete a guided adaptation demo

Module	Hours	Topics	Practical outcome
12. Evaluation, Reliability, and Responsible Use	6	Test-set design; correctness; groundedness; retrieval relevance; citation accuracy; tool-call success; human review; limitations of LLM judges; prompt injection; privacy; bias; regression testing	Build an evaluation suite containing normal, edge, and adversarial cases
13. Deployment and Application Operations	4	User interface and API basics; secrets management; access control; logging and tracing; latency; caching; usage budgets; deployment; maintenance and feedback	Deploy a small application with basic monitoring
14. Capstone Integration and Presentation	6	Requirements review; architecture; integration; testing; documentation; demonstration; individual viva	Present a working application and its evaluation results
Total	**90**		

Practical projects

1. **Prompt-based productivity assistant**
   Generate summaries, professional emails, and structured information from sample business content. Compare results across a fixed set of test cases.
2. **Knowledge-base assistant using RAG**
   Answer questions from approved PDFs or course documents, provide citations, and clearly indicate when the documents do not contain an answer.
3. **AI agent with tools**
   Retrieve course information, check a sample timetable, and prepare an enquiry record. Require confirmation before any external write or communication.

4. **Final capstone — choose one**

Project	Core functionality
Cypher Analytica AI Course Counsellor	Explain approved course details, compare options, answer FAQs, and prepare enquiry records
LMS AI Tutor	Answer syllabus questions, provide hints, generate practice questions, and cite learning materials
Research Document Assistant	Compare uploaded papers, extract study details, and produce source-linked summaries
Business Knowledge Assistant	Search internal policies and manuals with access restrictions and evidence-backed answers

5. Each submission should include source code, setup instructions, architecture, a demonstration, a test report, and documented limitations.

Assessment and certification

Assessment	Weight
Module assignments and lab work	30%
Concept quizzes	10%
Practical examination	20%
Capstone project	30%
Individual demonstration and viva	10%
Total	**100%**

Proposed certification criteria: minimum 75% attendance, 60% overall marks, and a functioning capstone with an individual demonstration.